From Nan obits to Galaxies: The Thermodynamics of Information as the Foundation of the HIGM Framework

Samir A Hamouda

Department of Physics, University of Benghazi, Benghazi, Libya

E:mail:dr_s_hamouda@yahoo.ie

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(This model was developed in the framework of theoretical physics and AI)

Abstract:

This article establishes the theoretical bridge between the proven principles of **Nanoscale Entropy Production** and the macroscopic gravitational framework of the **Hamouda Informational Gravity Model (HIGM)**. Recent experimental breakthroughs in nanoscale thermodynamics have confirmed that information processing—specifically the erasure and manipulation of bits—produces a measurable entropic force. This paper explores the transition of these microscopic laws to cosmological scales through the **Hamouda Informational Gravity Model (HIGM)**. We demonstrate that the **Infrared Information Flux** (Φ_{IR}) serves as the macroscopic analogue to nanoscale dissipation, providing a welcoming theoretical landscape for gravity as an emergent entropic force.

Keywords:

- **Primary Physics:** HIGM (Hamouda Informational Gravity Model), Informational Vacuum, Emergent Gravity, Entropic Force, Infrared Information Flux (Φ_{IR}) .
- **Information Theory:** Shannon Entropy ($S_{Shannon}$), Landauer's Principle, Quantum Information Substrate, Bit Erasure, Informational Impedance (H).
- **Nanoscale Thermodynamics:** Nanoscale Entropy Production, Non-Equilibrium Systems, Dissipative Structures, Thermal Gradients.
- **Cosmology & Astrophysics:** Hubble Tension Resolution, Entropy Friction, Gravitational Latency, Dark Matter Alternative, Large-Scale Structure.

- **Experimental Validation:** LIGO-Virgo-KAGRA Residuals, JWST Early Galaxy Monsters, Galaxy Cluster Scaling

1. Introduction

For decades, the "Information Paradox" was confined to black hole event horizons and theoretical thought experiments. However, recent advances in **nanoscale physics** have moved information theory into the laboratory. Experimental verification of **Landauer's Principle** has shown that "Information is Physical."

The HIGM takes these verified microscopic principles and applies them to the vacuum of space. It posits that the universe is not an empty stage but a dense **Informational Vacuum**. Just as Brownian motion in a fluid emerges from microscopic collisions, the HIGM proposes that gravity emerges from the macroscopic gradient of information entropy produced by the flux of infrared radiation.

2. The Mathematical Bridge

2.1. The Nanoscale Foundation: Landauer's Principle

At the nanoscale, erasing one bit of information produces a minimum amount of heat (Q):

$$Q \geq k_B T \ln 2$$

This creates an **Entropic Force** (F_e) acting on the system:

$$F_e = T \nabla S_{info}$$

Where ∇S_{info} is the gradient of information entropy.

2.2. The Macroscopic Transition: The HIGM Equation

The HIGM elevates this to the cosmic scale by replacing local thermal gradients with the **Infrared Information Flux** (Φ_{IR}). The emergent gravitational acceleration (g) is derived from the informational saturation of the vacuum (H):

$$g_{HIGM} \approx \Phi_{IR} \cdot \frac{S_{Shannon}}{H}$$

By integrating the 9.05% informational boost verified in galactic rotation curves, the model bridges the gap between the quantum "bit" and the galactic "cluster."

3. Discussion: Why the Nanoscale Welcomes HIGM

The success of HIGM is supported by three key behaviors observed at the nanoscale:

1. **Non-Equilibrium Stability:** Just as Nano machines operate in "noisy" environments, HIGM explains galactic stability through the "noise" of the **Informational Vacuum**.
2. **Impedance and Drag:** Our derived **"Entropy Friction"** (the 5μs delay in gravitational waves) is the macroscopic version of the informational drag observed in high-speed Nano processors.
3. **Phase Transitions:** The "Big Bang Reboot" in HIGM matches the phase transitions seen when information density reaches a critical threshold in quantum computing substrates.

4. Conclusion

The HIGM model is not an isolated theory; it is the logical extension of **21st-century thermodynamics**. By moving from a "Material-based" to an **"Information-based"** paradigm, we resolve the Hubble Tension and the Dark Matter crisis using the same laws that govern the smallest chips in our pockets. The universe, in the HIGM view, is a self-organizing informational processor, where gravity is simply the "overhead" of cosmic data flow.

5. Key References

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